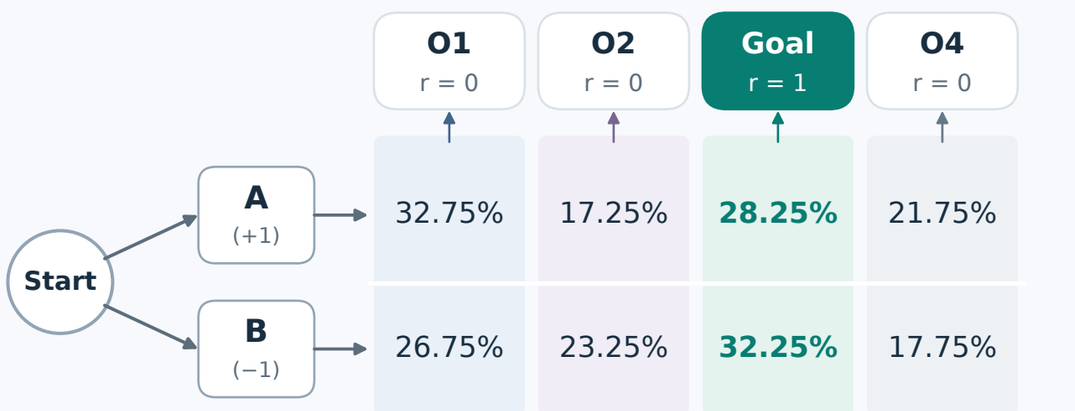
A squared-loss optimum can prefer the wrong action

Two-dimensional features · Natural uniform reference · Spherical backward features · Different probability classes

ONE-DECISION MDP Two equally likely starts · Two actions each · Four terminal outcomes · γ = 0 · Reward on arrival

Initial configuration 1

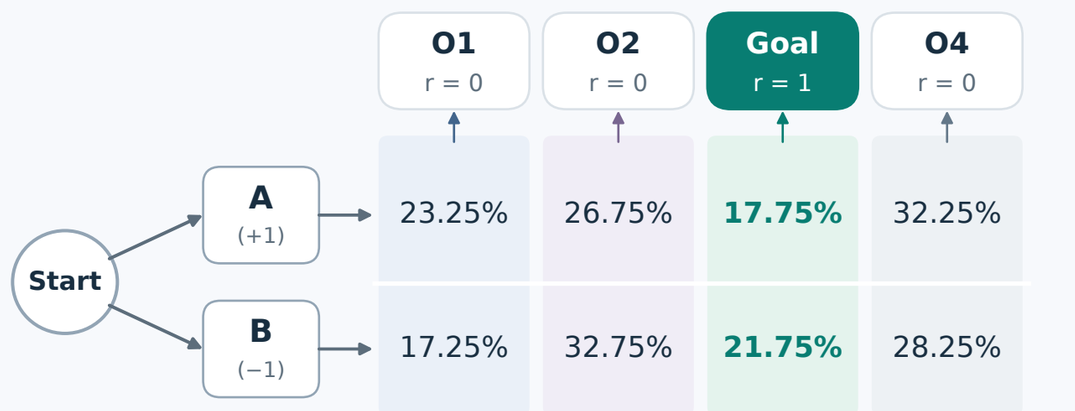
x = +1



Choose an action → sample one terminal outcome

Initial configuration 2

x = −1



Choose an action → sample one terminal outcome

The same four terminal states appear in both panels. Training samples all four start–action pairs equally.

What each value function chooses

Goal value = probability of reaching Goal

	Configuration 1 value(A) / value(B)	Configuration 2 value(A) / value(B)	Actual success under true transitions
True values Optimal action under the MDP	0.2825 / 0.3225 Δ = −0.0400 → **choose B**	0.1775 / 0.2175 Δ = −0.0400 → **choose B**	**27%** average of the two starts
Optimal FB values Unique global squared-loss prediction	0.3050 / 0.2950 Δ = +0.0100 → **choose A**	0.2050 / 0.1950 Δ = +0.0100 → **choose A**	**23%** average of the two starts
Explicit density model Feasible construction; no optimum computed	0.27755 / 0.32745 Δ = −0.0499 → **choose B**	0.18245 / 0.21255 Δ = −0.0301 → **choose B**	**27%** average of the two starts

Δ = value(A) − value(B). Actions maximize the displayed value; success is always evaluated in the true MDP.

Certificate: every global cross-entropy optimum chooses B in both configurations

Feasible mean KL **0.000206499** < **0.000474209** KL lower bound for any wrong or tied decision

Goal / non-Goal coarsening gives the bound; a global optimum cannot have more KL than a feasible model.

Model geometry also changes: squared loss on the softmax class succeeds too. This figure does not isolate cross-entropy.

A two-feature FB model can preserve the exact action differences for every reward.

Finite-capacity population construction · Exact reward integration · No learned actor or reward-inference evaluation